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[22BSCSMA51]

FIFTH SEMESTER B.Tech EXAMINATION, DECECEMBER-2024

INFERENTIAL STATISTICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: **(i) Missing data may be suitably assumed**

(ii) Answer all Questions

		PART-A	5 x2=10		
1.	a.	Describe primary data and secondary data with suitable examples	L2	CO1	02
	b.	The variance of a population is 2. The size of the sample collected from the population is 144. What is the standard error (S.E.) of mean?	L2	CO2	02
	c.	State atleast two applications of Student's t- distribution.	L2	CO3	02
	d.	Outline when to use non-parametric tests in statistical analysis?	L2	CO4	02
	e.	If $n = 100, \sigma = 5$, determine the maximum error with 95% confidence. ($Z_{0.05} = 1.96$)	L3	CO5	02
		PART-B	6X5=30		
2.	a.	Explain the probabilistic sampling methods with suitable examples.	L2	CO1	05
		OR			
	b.	A random sample of size 100 is taken from an infinite population having the mean $\mu = 76$ and the variance $\sigma^2 = 256$. What is the probability that $\bar{x}$ will be between 75 and 78. ($A(0.625) = 0.2334$ & $A(1.25) = 0.3944$)	L3	CO1	05
3.	a.	Explain one-tailed test and two-tailed test in detail.	L2	CO2	05
		OR			
	b.	The average breaking strength of the steel rods is specified to be 18.5 thousand pounds. To test this, a sample of 14 rods were tested. The mean and standard deviations obtained were 17.85 and 1.955 respectively. Is the result of experiment significant? ($t_{0.025,13} = 2.160$)	L3	CO2	05
4.	a.	An oceanographer wants to check whether the mean depth of the ocean in a certain region is 57.4 fathoms with a standard deviation of 5.2 fathoms, as had previously been recorded. What can he conclude at level of significance α = 0.05, if readings taken at 40 random locations in the given region yielded a mean of 59.1 fathoms. ($Z_{0.05} = 1.96$)	L3	CO3	05
		OR			
	b.	A manufacturer claims that 20% of their products are of top quality. However, in a production run of 400 articles, only 50 were found to be of top quality. Test the hypothesis that the proportion of top-quality products differs from the manufacturer's claim at 5% level of significance. ($Z_{0.05} = 1.96$)	L3	CO3	05

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5.	a.	Determine whether the following sequence is random: F F F M M F F F F M F M M M F F F F M M F F F M M Test at 5% level of significance. ($R_{0.05,\ (15,10)} = (7,18)$)	L3	CO4	**05**
		OR			
	b.	Following are the responses to the question "How many hours do you study before a major statistics test?" 6 5 1 2 2 5 7 5 3 7 4 7 Use Sign test to test the hypothesis at 5% LOS that the median number of hours a student studies before a test is 3. ($T_{0.05,11} = 1$)	L3	CO4	**05**
6.	a.	Explain the properties of good estimators.	L2	CO5	5
		OR			
	b.	The mean and standard deviation of a population are 11,795 and 14,054 respectively. What can one assert with 95% confidence about the maximum error if $\bar{x} = 11{,}795$ and n = 50, also construct 95% confidence interval for the true mean. ($Z_{0.05} = 1.96$)	L3	CO5	5
7.	a.	Discuss the following: **i.** Point Estimation **ii.** Interval Estimation	L2	CO5	5
		OR			
	b.	Consider the following sample data representing the scores of 6 students on a test: $X = \{85, 90, 95, 80, 70, 100\}$. Estimate the mean and variance using Maximum Likelihood Estimation (MLE).	L3	CO6	5
		PART-C	**6 X 10=60**		
8.	a.	Explain the census method and the sampling method in detail. How do they differ from each other?	L2	CO1	**10**
		OR			
	b.	Samples of size 2 are taken from the population 1, 2, 3, 4, 5, and 6 without replacement. Determine: **i.** The mean of the population. **ii.** The Standard deviation of the population. **iii.** The mean of the sampling distribution of means. **iv.** The standard deviation of the sampling distribution of means.	L3	CO1	**10**
9.	a.	Producer of a particular branded cigarette claims that the nicotine content in his cigarette on the average is 1.83mg. Can this claim accepted if a random sample of 8 ' cigarettes' of this type have the nicotine contents of 2.0, 1.7, 2.1, 1.9, 2.2, 2.1, 2.0, 1.6mg? Use a 0.05 level of significance. ($t_{0.025,7} = 2.365$)	L3	CO2	**10**
		OR			
	b.	To compare two kinds of bumper guards, 6 of each kind were	L3	CO2	**10**

mounted on a car and then the car was run into a concrete wall. The following are the costs of repairs:

Guard 1	107	148	123	165	102	119
Guard 2	134	115	112	151	133	129

Use the 0.01 level of significance to test whether the difference between two sample means is significant. ($t_{0.005,10} = 3.169$)

10.	**a.**	Two types of new cars produced in U.S.A. are tested for petrol mileage, one sample is consisting of 42 cars averaged 15 kmpl while the other sample consisting of 80 cars averaged 11.5 kmpl with population variances as ${\sigma_1}^2 = 2.0$ and ${\sigma_2}^2 = 1.5$ respectively. Test whether there is any significance difference in the petrol consumption of these two types of cars at 1% LOS. ($Z_{0.01} = 2.58$)	L3	CO3	**10**
		OR			
	b.	Random samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 200 men and 325 women were in favour of the proposal. Test the hypothesis that proportions of men and women in favour of the proposal are same, at 5% level. ($Z_{0.05} = 1.96$)	L3	CO3	**10**
11.	**a.**	Determine if the following data is collected at random: 18, 36, 19, 22, 25, 44, 23, 27, 27, 35, 19, 43, 37, 32, 28, 43, 46, 19, 20, 22. Test at 5% level of significance using Runs test. ($R_{0.05,\ (9,9)} = (5,15)$)	L4	CO4	**10**
		OR			
	b.	A teacher wants to determine if the median number of hours students study before a major statistics test is different from 3 hours. A sample of 10 students is surveyed, and their responses (in hours) are as follows: 2.5 2 4 4.5 4 2.5 4.5 3 3.5 5 Use the Single Sample Wilcoxon Signed-Rank Test to test the hypothesis at a 5% significance level. ($W_{0.05,\ 9} = 5$)	L4	CO4	**10**
12.	**a.**	For the following data, test the hypothesis that the median HDL cholesterol levels in adult population of city A and city B are the same using Mann-Whitney U test at 5% LOS ($U_{0.05,(7,5)} = 5$): City A: 42 20 51 39 57 60 23 City B: 30 42 25 29 35	L4	CO4	**10**
		OR			
	b.	Company trainees are randomly divided into three groups of 11 each and are given a course of management skill by three different methods. At the end of the training period, they are given a test and their scores are as follows: Method A: 99 64 101 85 79 88 97 95 90 100 Method B: 83 102 125 61 91 96 94 89 93 75 Method C: 89 98 56 105 87 90 87 101 76 89	L4	CO4	**10**

		Use Kruskal-Wallis test to determine that three methods are equally effective at 5% LOS. ($\chi^2_{0.05,2}$=5.99)			
13.	**a.**	The weight of ball bearings proved in a certain factory are normally distributed with mean 0.5 Newton's and S.D 0.002 Newton's. Find the percentage of ball bearings produced with weights **i.** less than 0.498 Newton's **ii.** more than 0.501 Newton's **iii.** between 0.4995 and 0.5005 Newton's $(A(1) = 0.3413, A(0.5) = 0.1915, \& A(0.25) = 0.0987)$	L3	CO5	**10**
		OR			
	b.	A researcher is studying student attitudes toward co-education. An attitude scale was administered to a sample of 240 students, who were asked to categorize their responses as favourable, neutral, or unfavourable. The results of the survey are summarized in the following table: **Response Category** — **Frequency** Favourable — 70 Neutral — 50 Unfavourable — 120 Use Chi-Square goodness of fit test to determine if there is a significant difference in attitudes toward co-education based on the collected responses at 5% LOS. ($\chi^2_{0.05,\,2} = 5.99$)	L3	CO6	**10**